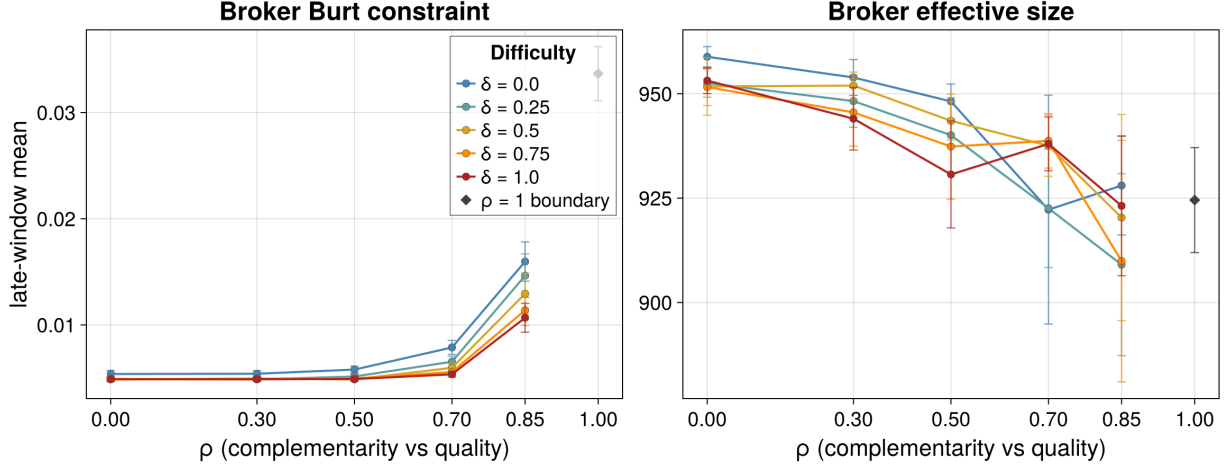


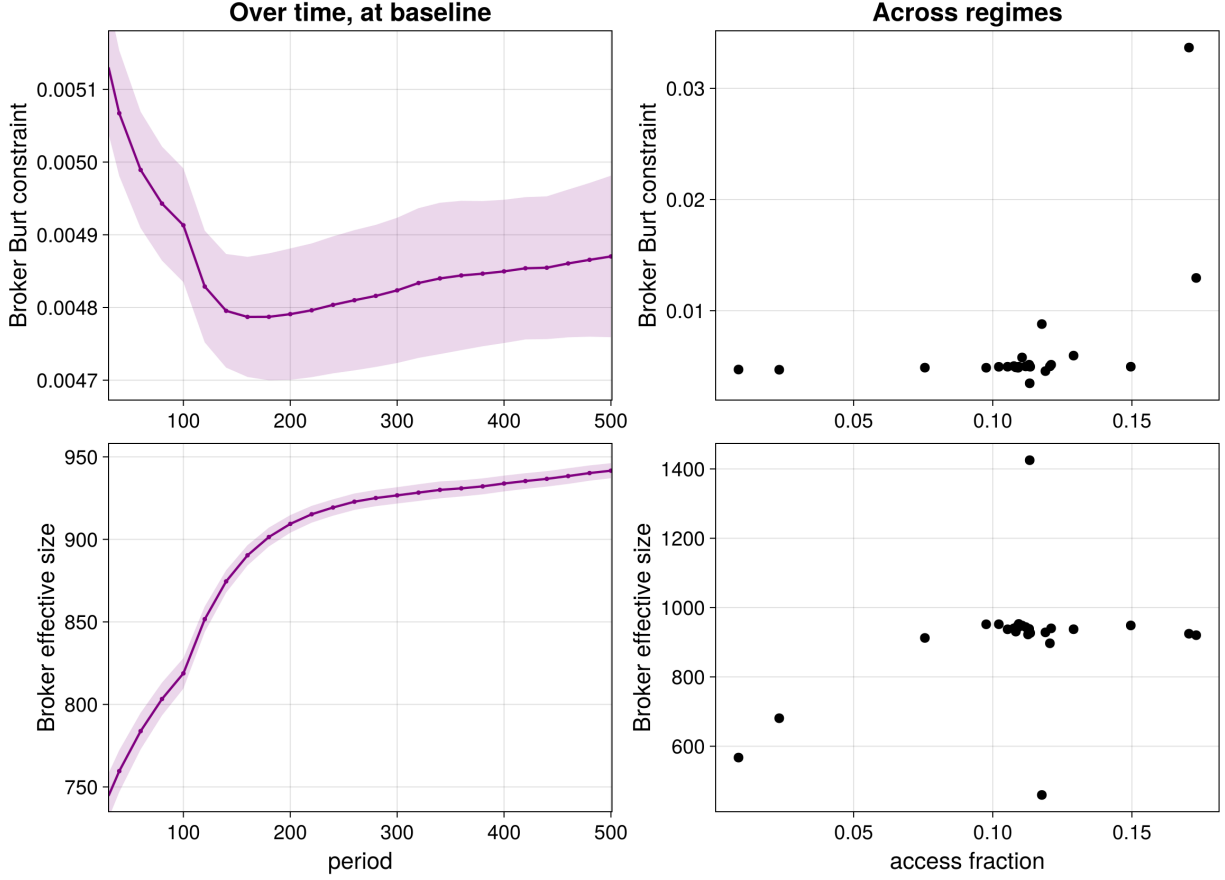
## Supplementary Material

### Alternative Structural Measures

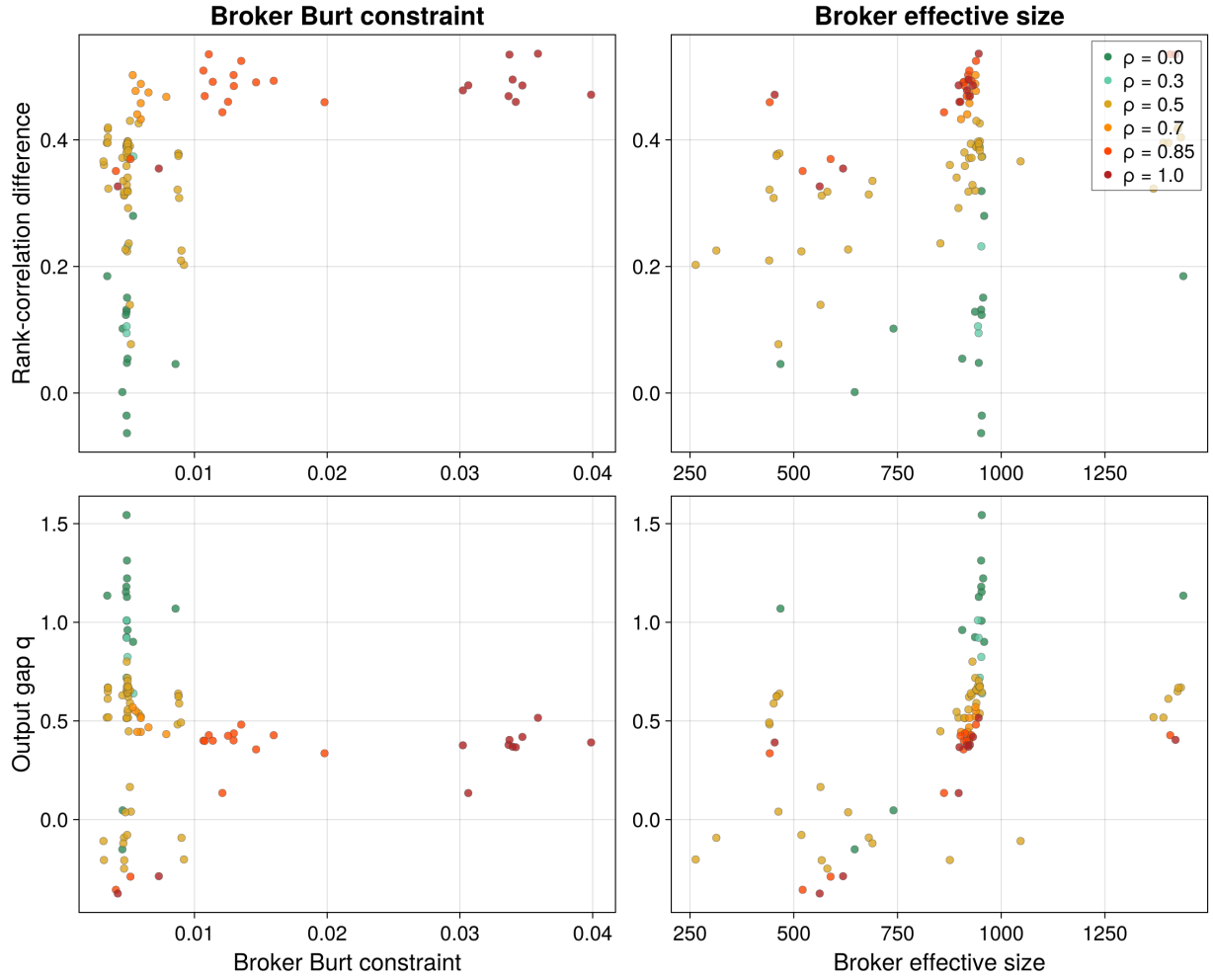
The first three figures reproduce the main structural-advantage analyses with the broker's two other saved ego-network measures, Burt's aggregate network constraint and Burt's effective size. Both are recomputed for the broker every 20 periods. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.



**Figure S1:** Broker Burt constraint (left) and broker effective size (right) across the  $\rho \times \delta$  grid. Lines for each difficulty level  $\delta$  span  $\rho = 0$  through 0.85. The pure-quality boundary  $\rho = 1$ , where  $\delta$  is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.



**Figure S2:** Broker Burt constraint (top) and broker effective size (bottom). Left: the measure over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals, measured every 20 periods. Right: one dot per one-at-a-time regime, the measure against access fraction, each a late-window mean over that regime's seeds. Time starts at  $t = 30$  to omit the initialization transient.



**Figure S3:** Difference in holdout rank correlation (broker minus principal, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right); one dot per regime, each a late-window mean over that regime's seeds, colored by the matching parameter  $\rho$ .